

2 FEASIBILITY STUDY FOR HIGHWALL MINING

2.1 Summary

This chapter presents two practical tools for highwall mining feasibility study, i.e. the Highwall Mining Index (HMI) for CHM mining and the Auger Mining Index (AMI) for augering. The tools were designed to help mine managers and mining engineers in evaluating the feasibility of highwall mining in a given pit. Minimum conditions required for highwall mining are also discussed prior to the presentation of the tools.

2.2 Introduction

Before employing highwall mining methods, two basic questions need to be answered:

- ❑ Is it *possible* for the pit to be mined using highwall mining methods?
- ❑ Is it *feasible and profitable* to mine the pit using highwall mining methods?

To answer the first question, one needs to examine the pit against the *minimum conditions* for highwall mining. To answer the second question as to feasibility, it is suggested that the *highwall mining index* for CHM or the *auger mining index* for augering, be used to evaluate the overall condition of the pit. In general, economic issues are not the concern of this report, although many of the feasibility issues have economic implications. The following sections present the minimum conditions and the highwall mining index and auger mining index for feasibility study.

2.3 Minimum conditions for highwall mining.

There are some circumstances where highwall mining may not be feasible for safety, productivity, profitability or environmental reasons. Based on the experience of highwall mining in Australia over the past 9 years, the following conditions are almost certain to preclude the use of highwall mining:

- ❑ **Poor roof condition.** Roof falls reduce mining progress, and can damage or bury mining equipment. If poor roof conditions are encountered, the site cannot be highwall mined using CHM. In general, auger mining is not sensitive to roof conditions. Even in extremely soft roof and floor conditions, leaving some coal in the roof and floor may facilitate successful mining operations. Conditions preventing highwall mining using CHM include: extremely weak to very weak rock; laminated rock, closely spaced rock jointing, major structures such as thrust faults close to the coal/roof interface or extremely weathered rock.
- ❑ **Limited mining reserve.** If the mineable reserve is too small, then the cost of equipment mobilisation and pit preparation etc. is likely to be prohibitive. The minimum tonnage of the reserve depends on the availability of the mining equipment on site. As a general guide, for a new mine to use CHM highwall mining, a 250 m mineable pit length is considered to be the minimum. For a mine with existing highwall mining operations, a 100 m pit length is the minimum for CHM, and a 50 m pit length is the minimum for auger mining.

- ❑ ***Seam methane content > 10 m³/tonne.*** As a general guide, a seam with methane content > 10 m³/tonne is considered to be unsuitable for highwall mining if inertisation of the entries is not possible or economically feasible.
- ❑ ***Seam dip >20°.*** Existing CHM systems can mine coal seams dipping up to 15 degrees downward (BHP steep-dip machine at the Moura Mine). The commonly used Addcar Systems can mine up to 10 degrees downward for a penetration of 300 m, or up to 15 degrees for lesser penetration. The Archveyor system has been used to mine up to 18 degrees at the Moura Mine. However, Nova Construction, Canada, have built a Highwall Miner that is designed to mine up to a 26 degree dip. Auger systems can mine down dips of up to 20 degrees. A seam with a dip greater than 20 degrees cannot be mined using any existing highwall mining system in Australia without modification.
- ❑ ***Unstable highwall and/or lowwall.*** Highwall mining operations are controlled from the launch pad immediately beside the highwall. A grossly unstable highwall, or a large scale lowwall instability in combination with a narrow pit, will preclude highwall mining operations in that part of the pit.
- ❑ ***Seam thickness <1.0 m.*** A seam less than 1.5 m thick cannot be mined using CHM, and seams less than 1.0 m cannot be mined economically using an auger.
- ❑ ***Pit width <30 m.*** A minimum pit width of 30 m is required to mobilise and operate the highwall mining system.
- ❑ ***Stone bands in coal seam.*** Hard stone bands thicker than 0.2 m in the coal seam may prevent using an auger system. In addition, seams containing dirt bands or clay seams sensitive to moisture occurring any where across the hole diameter are difficult to mine using an auger. CHM mining can tolerate stone bands although incendiary sparking may be an issue.

If any of the above conditions are encountered at the given site, the site is likely to be rejected for further consideration for highwall mining. If however, none of these adverse conditions are present, a second evaluation should be carried out to quantify the suitability of the site for highwall mining and the likelihood of success. The highwall mining index (HMI) for CHM and the Auger Mining Index (AMI), discussed below, provide tools for this purpose.

2.4 Highwall mining index (HMI) for CHM

The suitability of a site for highwall mining depends on a number of geological, geotechnical, operational and commercial parameters. In particular, the parameters listed above must individually satisfy the minimum requirement for highwall mining (i.e. pass step 1), but they may have different importance in the overall assessment. For this reason, all relevant parameters need to be assessed systematically, since some combinations of parameters may still threaten a safe and profitable mining operation. A practical way of doing a systematic assessment is to quantify the condition of each parameter and project the scores of all the parameters into one index value. This index value will reflect the overall condition of the site for highwall mining, and can help in deciding whether to employ highwall mining at this site. The highwall mining index (HMI) method was proposed by W. Seib (1992). It has been modified for this report based on the experience of the last nine years of highwall mining practice in Australia.

The HMI focuses on CHM operations, which are the most widely used in the industry. Where geotechnical conditions do not permit the application of CHM, an auger system may be considered (see Section **Error! Reference source not found..**)

It is emphasised that the HMI system is intended as the second step of the initial evaluation, not as the definitive assessment. It is also emphasised that there will always be uncertainty arising from unexpected geological structures or other variability in conditions which may be encountered during mining operations.

The user of this method requires an adequate understanding of all of the parameters listed in the next section.

2.4.1 Key parameters

The 15 key parameters, which have had a major effect on the success of highwall mining, including the 8 parameters listed in section 2.3, are summarised and grouped below in four categories based on their impact on decision-making. Guidelines for the assignment of numerical values to these parameters are given in Table 2.1.

Group A – Most critical parameter. The roof condition is the single most critical parameter for highwall mining using CHM. Many of the unsuccessful highwall mining operations in the past were a direct or indirect result of poor roof conditions.

A Roof condition – Roof falls can cause frequent withdrawal of the mining system and shut down the mining process. In the worst case, they can damage or bury the CHM system. Roof falls are often the major cause of limited penetration. Experience has shown that good roof is one of the most important parameters for successful highwall mining.

Group B – Critical parameters. This group of parameters are considered critical in deciding whether highwall mining operations will be possible in a given pit.

B1 Mineable reserve – If the reserve is small, the fixed costs such as pit preparation and mobilisation will be a much larger proportion of the total cost and mining therefore may not be economically feasible.

B2 Seam gas – Seam gas directly affects the safety of highwall mining operations, as for underground mining. If the seam has a very high methane content, the seam simply cannot be mined using current highwall mining systems as it is impossible to dilute and ventilate the methane adequately.

B3 Seam dip – If the seam dip is greater than 15 degrees downward, the seam cannot be mined with systems currently in use in Australia. Gently dipping coal seams may be mined. Very limited experience of mining upward inclined seams is available. But, an upward slope greater than 5 degrees is considered to be unsuitable for highwall mining.

- B4 Highwall/lowwall stability** – Since highwall mining systems operate at the foot of the highwall and the lowwall, an unstable highwall or lowwall is a serious threat to safety. No operations will be viable under such conditions. For small scale rock falls from the highwall or lowwall, protective measures need to be implemented.

Group C – Sub-critical parameters. Parameters in this group significantly affect productivity and profitability. Although they are not significant enough to preclude employing highwall mining, they often dominate the economic feasibility of using this mining method.

- C1 Seam continuity** – Faulting that displaces the seam by more than 1 m often stops highwall mining, and hence significantly limits the penetration distance and resource recovery.
- C2 Pit condition** – Highwall mining requires adequate space and a clean floor in the pit for equipment movement and storage. If the pit is too narrow, or the floor is too soft and muddy, excavation of the lowwall or digging into the floor may be required. This may not be geotechnically or economically feasible.
- C3 Seam quality** – This parameter will directly affect the profitability of highwall mining. Depending on the market price, a low quality coal seam may provide little profit margin. Low quality could be a result of low coal rank and/or dirt bands in the coal seam, which dilute the overall coal quality. Highwall mining can improve product quality by selectively mining the higher quality section of the seam.
- C4 Floor condition** – Soft floor conditions cause the continuous miner to trip and hence slow or even stop mining. They can also undermine pillar/panel stability.
- C5 Ground water** – Excessive water flow into a mining entry may force the withdrawal of the system and termination of the entry.

Group D - Important parameters. Parameters in this group are non-critical parameters to decision making but they can have a substantial impact on the productivity and profitability of the operation.

- D1 Coal strength** – This parameter will affect the resource recovery since it will affect the pillar size.
- D2 Cover depth** – This parameter affects the pillar size required to maintain the panel stability.
- D3 Subsidence** – In the long term, subsidence may occur in many highwall mined pits. If long term stability is required, measures such as back-filling or leaving large pillars, may be necessary. Back-filling entries will increase operating costs, whereas using oversized pillars will reduce the resource recovery rate.
- D4 Seam thickness** – The mineable seam thickness for single-pass CHM ranges from 1.5 m to 4 m. Optimal seam thickness helps to achieve the best productivity and resource recovery rate. It is also desirable that seam thickness is uniform within a reserve.

D5 Mobilisation and service – Local or onsite availability of the mining system and service could affect the mining productivity and the operating cost.

2.5 Conditions of the key parameters

Each of the 15 key parameters is categorised into four conditions for a given pit: Excellent, Good, Fair and Poor. The HMI user assigns a rating in the range of 0-100 to each parameter based on its condition:

Excellent:	80-100,	default = 90
Good:	60-80,	default = 70
Fair:	40-60,	default = 50
Poor:	0-40,	default = 20

Where there is insufficient information to make a judgement a default value is given to each parameter.

A description of the different conditions for the 15 key parameters is given in Table 2.1.

Table 2.1. Rating guidelines for the 15 key parameters included in the highwall mining index.

Rank	Parameter	Condition				
		Excellent (100≥rating>80)	Good (80≥rating>60)	Fair (60≥rating>40)	Poor (40≥rating>0)	No-go (rating=0)
A	Roof condition	Strong to very strong roof rocks; thickly bedded; joints with wide spacing. No alternation and weathering.	Moderately strong roof rock; medium bedded; joints with moderate spacing. Slight alternation and weathering.	Moderately weak roof rocks; thinly bedded; joints with close spacing. Alternation and weathering.	Weak roof rocks; very thinly bedded; joints with very close spacing. Strong alternation and weathering.	Extremely weak to very weak rock; laminated; joints with very close spacing. Severe weathering
B1	Mineable Reserves	>1500 m highwall	500 m to 1500 m highwall	250 m to 500 m highwall	100 m to 250 m highwall	<100 m highwall
B2	Seam gas (methane)	Very low methane content (<0.1m³/t). No concern to mining	Low methane content in general. (0.1-1m³/t). Higher gas only in localised pockets.	Seam methane = 1-5m³/t. Gas problems could be encountered occasionally in reserve.	Seam gas (5-10m³/t). Gassy seam and gas problems are expected to be encountered throughout.	Seam methane >10m³/t
B3	Seam Dip	+0° to -4°	+0° to +1°, or -4° to -7°	+1° to +2°, or -7° to -10°	+2° to +4°, or -10° to -15°	>+4°, <-15°
B4	Highwall/low wall stability	Smooth highwall wall. No hanging blocks. No major joint sets or faults. No risk of large scale lowwall instability	Localised loose rock blocks. Slight weathering, Minor joints/cracks. No discernible risk of lowwall instability	Common small rock falls. Dense jointing. No risk of large scale highwall instability. Very low risk of lowwall instability. Possible block falls.	Large rock falls. Weathered. Wedges and slabs formed by joints. Some risk of local lowwall instability but manageable.	Grossly unstable highwall. Significant risk of large scale lowwall instability.
C1	Seam continuity	Seam is consistent with no faulting or folds	Structures which slightly reduce seam thickness. Only small faults and undulations	Structures which reduce thickness by up to 50%. Seam changes direction by 5° in folds	Structures which reduce thickness over 50%. Faults and distinct folds are common.	Seam heavily affected by faulting and folding.
C2	Pit condition, preparation	No loose material, dry, no floor work.	<0.5 m loose material. Water puddles. Shave floor.	<2 m loose material. Pumping required. Some mud. Soft floor.	>2 m loose material. Major pumping and mud clean up. Floor repairs required.	Pit width <30 m.
C3	Seam quality	Medium to hard export coking coal. No in-seam parting or stone bands	Soft coking coal and export steaming coal. 1 or 2 stone bands.	Any unwashed product which is saleable. Many partings and stone bands.	Coal not in previous categories which must be washed for sale. Many stone bands.	
C4	Floor condition	Strong floor rocks. Unaffected by water.	Weak to moderately strong floor rocks. May	Weak floor rocks. Occasionally muddy.	Very weak floor rocks (clay). Wet and muddy throughout.	

Rank	Parameter	Condition				
		Excellent (100≥rating>80)	Good (80≥rating>60)	Fair (60≥rating>40)	Poor (40≥rating>0)	No-go (rating=0)
			be wet but not muddy.			
C5	Ground water	Dry environment. No water gets into the mining entries.	Seasonal dry and wet environment. Some water into entries but is manageable.	Generally wet environment. Some water problems could be expected at fault zones.	Very wet environment. Water gets into entries from the pit and from the ground. It may flood the entry.	
D1	Coal strength	Lab coal UCS > 30MPa. Limited coal joints with wide spacing.	Lab coal UCS=20-30MPa. Coal joints with moderate spacing.	Lab coal UCS=10-20MPa. Coal joints with close spacing.	Lab coal UCS <10MPa. Heavily cleated. Coal joints with very close spacing.	
D2	Cover depth (overburden + spoil)	Overburden depth <80 m in the reserve	Overburden depth 50 – 100 m.	Overburden depth 60 – 120 m, possible stress concentration near highwall	Overburden depth 80 – 150 m, possible significant stress concentration near highwall	
D3	Subsidence	Surface subsidence not an issue.	Minor subsidence effects but all contained on lease. Minimal rehabilitation efforts.	Some subsidence or water effects off site but controllable. Rehabilitation could be costly.	No subsidence is allowed. Significant water effects off site. Major rehabilitation costs. Subsidence control is critical.	
D4	Seam thickness	>3.5 m	2.5 m- 3.5 m	2.0 m – 2.5 m	1.5 m - 2.0 m	<1.5 m.
D5	Mobilisation and service	Equipment, service capacity and skilled mining crew are available on site.	Equipment, service capacity and skilled mining crew are available in mines nearby.	Equipment, service capacity and skilled mining crew are available interstate.	Equipment and service capacity are available overseas. New mining crews need to be trained.	

* Terminology follows the ISRM standard.

2.5.1 Calculating the Highwall Mining Index (HMI)

The HMI is a mean value of the rating of each of the 15 parameters. The 15 parameters have been weighted according to their relative importance in the decision-making process. The individual weight factor of each parameter has been assigned according to the rank of the group that it belongs to. Parameters in the same group have been given with the same weight.

In practice, if a critical parameter is very poor (e.g. if the roof condition is extremely bad), then it will probably not be wise to proceed with highwall mining, even though all other parameters are excellent. The HMI reflects such extreme cases by using a weighted geometric mean value of the 15 parameters, rather than a weighted average value. The geometric mean value of the parameters will approach zero if one of the parameters is close to zero, whereas the average value would still be high even when the value of one of the parameters is close to zero.

The geometric mean value of the parameters in each of the four groups is given by:

$$\bar{A} = A$$

$$\bar{B} = (B1 \cdot B2 \cdot B3 \cdot B4)^{1/4}$$

$$\bar{C} = (C1 \cdot C2 \cdot C3 \cdot C4 \cdot C5)^{1/5}$$

$$\bar{D} = (D1 \cdot D2 \cdot D3 \cdot D4 \cdot D5)^{1/5}$$

In these expressions, each parameter in each group is given the same weight.

The HMI is the weighted mean value of the four groups:

$$HMI = (\bar{A}^{1.6} \cdot \bar{B}^{1.2} \cdot \bar{C}^{0.8} \cdot \bar{D}^{0.4})^{1/4}$$

The weighting assigned to each group is given by the exponential factors of 1.6, 1.2, 0.8 and 0.4 to groups A, B, C and D respectively.

The value of HMI ranges between 0 and 100. It represents the attractiveness of a site for highwall mining with a CHM. A score of 60 or greater indicates a prospect with a high potential for highwall mining, worthy of a detailed feasibility study. A score of less than 40 indicates a poor site, which would be unlikely to produce coal safely and profitably using the current methods of highwall mining using a CHM.

A score between 40 and 60 indicates that the site is not very attractive from a purely technical point of view. However, other non-technical issues, such as an urgent need for coal, or an increase in market price, may weight this site toward a potential mining site.

2.5.2 Confidence rating for the Highwall Mining Index

Assigning a rating (0-100) of the 15 key parameters in the HMI calculation is a subjective process. It is affected by the information available to the person who makes the judgement. Some parameters can be rated easily with full confidence whereas others can only be rated with a limited level of confidence. For instance, “pit condition” can be rated easily with a confidence level of near 100% since the pit and its floor are likely to be fully exposed and can be inspected visually. “Seam continuity”, however, may only be rated with a low confidence level as the understanding of local geology may not be sufficient, depending on the amount of information available.

New mines are likely to have a smaller data set and hence the confidence level in the values assigned to many parameters may be low. By contrast, proposed operations at an existing mining site may have detailed information available and so can assign higher confidence levels to their data.

Analysing the individual contributions to the confidence level of HMI will help to identify the parameters in need of further assessment.

To calculate the confidence level of HMI, each parameter is given a confidence level on a scale from 0 to 100%. The confidence level of an individual parameter is weighted in the same way as it is weighted in the HMI calculation. The total confidence level of the HMI is calculated by the weighted geometric mean of the confidence levels of each of the 15 parameters.

$$C_{\bar{A}} = C_A$$

$$C_{\bar{B}} = (C_{B1} \cdot C_{B2} \cdot C_{B3} \cdot C_{B4})^{1/4}$$

$$C_{\bar{C}} = (C_{C1} \cdot C_{C2} \cdot C_{C3} \cdot C_{C4} \cdot C_{C5})^{1/5}$$

$$C_{\bar{D}} = (C_{D1} \cdot C_{D2} \cdot C_{D3} \cdot C_{D4} \cdot C_{D5})^{1/5}$$

where C_x is the confidence level of parameter X.

The overall confidence level of HMI is given by:

$$C_{HMI} = (C_{\bar{A}}^{1.6} \cdot C_{\bar{B}}^{1.2} \cdot C_{\bar{C}}^{0.8} \cdot C_{\bar{D}}^{0.4})^{1/4}$$

A confidence score greater than 60% indicates that, in the opinion of the person making calculating the HMI, the calculated HMI is likely to be reasonably reliable. A score of less than 50% indicates a level of uncertainty in the evaluated HMI which needs to be resolved by getting additional information to better define the rating of those parameters for which there is currently greatest uncertainty.

2.5.3 Decision-making based on the HMI and its confidence value

For a given site, the calculated HMI and its confidence level are plotted as a point in the HMI-confidence level chart as shown in Figure 2.1. The chart area is divided into four regions and a transition zone. A subjective assessment is provided, depending on the region where the plotted point falls as shown in Figure 2.1.

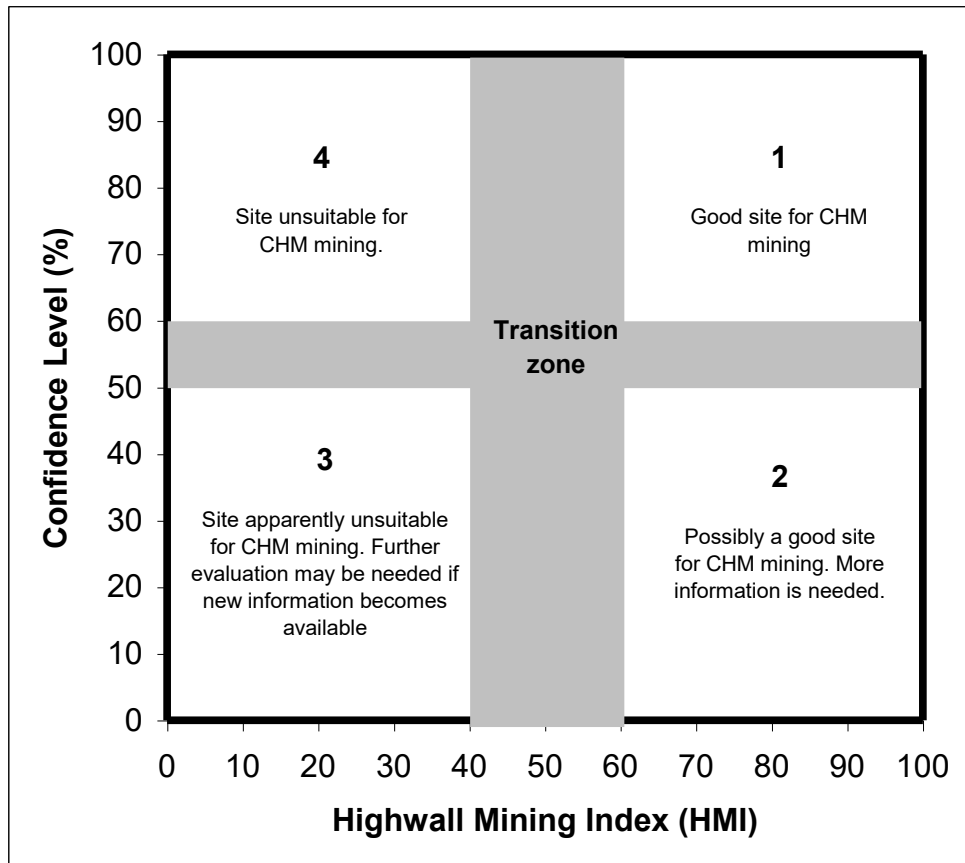


Figure 2.1. HMI – confidence level chart.

Region 1 – High HMI with high confidence level.

Based on the available information, the pit evaluated is a good potential site for highwall mining. Detailed mine planning and financial analysis should be carried out.

Region 2 - High HMI with low confidence level.

Based on the available information, the pit may be a good potential site for highwall mining, but significant uncertainty exists in the judgement rating of many parameters. The low confidence parameters need to be identified and carefully investigated. Some preliminary site investigation may be needed to reduce the uncertainty in these parameters.

Region 3 – Low HMI with low confidence level.

Based on the available information, the pit appears to have low potential for highwall mining. However, the low confidence in the parameter rating means that this assessment could change as more information becomes available.

Region 4 – Low HMI with high confidence level.

Based on the available information, the pit is not suitable for highwall mining. No further consideration of this pit for CHM mining is necessary. However, it may be suitable for auger mining

Transition zone – Medium HMI and medium confidence level.

Depending on where an assessment plots, a decision should be made as to whether to proceed to obtain more information. Other non-technical issues may also require consideration here.

2.6 Example cases

Case 1

A pit in Central Queensland was to be considered for highwall mining using CHM. Based on previous site investigations in neighbouring regions as well as observation of the current pit, the following information was available to the mining engineer:

Roof condition: Based on the observation of the highwall face, the immediate roof is thinly bedded weak mudstone, which has a thickness of about 0.5 m. It is overlain by a layer of stronger siltstone. The main roof overlying the siltstone is a thick layer of more competent, intact sandstone and interbedded siltstone. The immediate mudstone roof layer may create some roof falls, but the roof falls are not expected to extend through to the siltstone or sandstone layer above. Some joints exist in the roof but are not intense. **Rating:** 70 (good), **confidence level:** 70%.

Mineable reserve: The total length of the reserve is about 750 m along the highwall. There is no limitation to mining penetration, which could extend beyond 300 m. There are no other constraints, such as major faults. **Rating:** 80 (good), **confidence level:** 95%.

Seam gas: The seam has a low methane content. Previous highwall mining of the same seam in nearby areas did not experience major problems with seam gas. **Rating:** 75 (good), **confidence level:** 80%.

Seam dip: Based on information in the neighbouring areas, the seam dip is expected to be in the range of 4.5° to 8.0° down dip. **Rating:** 75 (good), **confidence level:** 70%.

Highwall stability: The final pit highwall was created using pre-split blasting techniques. The highwall face is generally smooth throughout the pit. No major faults exist which have the potential to create major highwall instability. Two joint sets were observed in some areas, which may create potential localised face instability. There are also some detached blocks from the highwall face and slumping of material from the highwall crest. There is no risk of lowwall failure. **Rating:** 75 (good), **confidence level:** 70%.

Seam continuity: Based on the observation of the final highwall and the previous strips, no major faults and folds are expected in the mining reserve. No in-seam drilling has been done. **Rating:** 90 (excellent), **confidence level:** 60%.

Pit condition: There is sufficient pit space (>60m) between the lowwall and highwall for highwall mining operations. The pit floor is competent with no soft material in general. Some rain water and mud has accumulated near the highwall toe. Pumping of the water is required, but this should be fairly easy using equipment already on site. **Rating:** 100 (excellent), **confidence level:** 100%.

Coal quality: The seam is good quality, low ash coking coal. No stone bands or partings were observed. **Rating:** 90 (excellent), **confidence level:** 95%.

Floor condition: The immediate floor is composed of carbonaceous mudstone up to 0.2 m thick, with a sheared clay layer within 0.5 m of the seam floor. The floor material is generally weak. The seam and the floor are likely to be wet. **Rating:** 60 (fair), **confidence level:** 50%.

Ground water: The mine site is generally dry and ground water is not a major concern. However, during the wet season, surface water may accumulate quickly and find its way to the mining entry from the pit floor or through geological structures. **Rating:** 70 (good), **confidence level:** 70%.

Coal strength: Extensive studies were conducted on coal strength of the seam in nearby areas. The seam is considered to be medium strong (UCS=20-30 MPa). The seam in this pit is well cleated and has regular joints. **Rating:** 75 (good), **confidence level:** 90%.

Cover depth: The cover depth is about 55 m at the highwall and is estimated to be around 90 m at the maximum penetration of 300 m. There is no spoil pile on the surface and the highwall is straight without any benches. No abnormal stress concentration is expected near the highwall. **Rating:** 80 (good), **confidence level:** 90%.

Subsidence: There is no concern about future subsidence of the mined ground. There is no water reserve which could be affected by any possible subsidence. **Rating:** 100 (excellent), **confidence level:** 95%.

Seam thickness: The seam at the highwall exposure has a thickness of 3.2 – 3.4 m. It is expected the thickness does not change significantly in the reserve. **Rating:** 80 (good), **confidence level:** 60%.

Mobilisation and service: The equipment, skilled mining crews and service capacity are on site. **Rating:** 90 (excellent), **confidence level:** 90%.

Using the above information, the process of calculating HMI is demonstrated below.

The average rating of the four groups:

$$\bar{A} = A = 70$$

$$\bar{B} = (B_1 \cdot B_2 \cdot B_3 \cdot B_4)^{1/4} = (80 \cdot 75 \cdot 75 \cdot 75)^{1/4} = 76$$

$$\bar{C} = (C_1 \cdot C_2 \cdot C_3 \cdot C_4 \cdot C_5)^{1/5} = (90 \cdot 100 \cdot 90 \cdot 60 \cdot 70)^{1/5} = 81$$

$$\bar{D} = (D_1 \cdot D_2 \cdot D_3 \cdot D_4 \cdot D_5)^{1/5} = (75 \cdot 80 \cdot 100 \cdot 80 \cdot 90)^{1/5} = 84$$

The highwall mining index (HMI):

$$HMI = (\bar{A}^{1.6} \cdot \bar{B}^{1.2} \cdot \bar{C}^{0.8} \cdot \bar{D}^{0.4})^{1/4} = (70^{1.6} \cdot 76^{1.2} \cdot 81^{0.8} \cdot 84^{0.4})^{1/4} = 75$$

The confidence level of each group:

$$C_{\bar{A}} = C_A = 0.7$$

$$C_{\bar{B}} = (C_{B_1} \cdot C_{B_2} \cdot C_{B_3} \cdot C_{B_4})^{1/4} = (0.7 \cdot 0.8 \cdot 0.7 \cdot 0.7)^{1/4} = 0.72$$

$$C_{\bar{C}} = (C_{C_1} \cdot C_{C_2} \cdot C_{C_3} \cdot C_{C_4} \cdot C_{C_5})^{1/5} = (0.6 \cdot 1.0 \cdot 0.95 \cdot 0.5 \cdot 0.7)^{1/5} = 0.72$$

$$C_{\bar{D}} = (C_{D_1} \cdot C_{D_2} \cdot C_{D_3} \cdot C_{D_4} \cdot C_{D_5})^{1/5} = (0.7 \cdot 0.9 \cdot 0.9 \cdot 0.95 \cdot 0.7 \cdot 0.9) = 0.86$$

The confidence level of HMI:

$$C_{HMI} = (C_{\bar{A}}^{1.6} \cdot C_{\bar{B}}^{1.2} \cdot C_{\bar{C}}^{0.8} \cdot C_{\bar{D}}^{0.4})^{1/4} = (0.7^{1.6} \cdot 0.72^{1.2} \cdot 0.72^{0.8} \cdot 0.86^{0.4})^{1/4} = 0.73 = 73\%$$

Table 2.1 Table 2.2 summarises the individual rating of each of the 15 parameters and the resultant HMI and its confidence level.

The results indicate that, for the pit evaluated, the HMI is 75 and the confidence level is 73%. When plotted on the HMI – Confidence chart, this pit falls into the centre of Region 1 (Figure 2.2). Hence, the pit is considered to be a good potential site for highwall mining. However, some parameters with low confidence levels may need to be examined in more detail before final mine planning and financial analysis is undertaken. These parameters include roof condition, seam continuity, floor condition, and seam thickness. Geotechnical investigations to better evaluate these parameters would be desirable before a final commitment to highwall mining.

Clearly, the sensitivity of the HMI to any of the parameters can be tested by changing one or more parameters and recalculating the HMI.

A highwall mining operation was eventually carried out in the pit, after a thorough site investigation and geotechnical design. The pit was mined successfully.

Table 2.2. Rating of the 15 key parameters and their confidence level, example pit 1.

Rank	Parameter	Judgement Rating	Confidence level (%)
A	Roof condition	70	70
B1	Mineable reserve	80	70
B2	Seam gas	75	80
B3	Seam dip	75	70
B4	Highwall stability	75	70
C1	Seam continuity	90	60
C2	Pit condition (pit preparation)	100	100
C3	Coal quality	90	95
C4	Floor condition	60	50
C5	Ground water	70	70
D1	Coal strength	75	90
D2	Cover depth	80	90
D3	Environmental aspect – subsidence	100	95
D4	Seam thickness	80	70
D5	Mobilisation and services	90	90
		Highwall Mining Index (HMI) = 75 Confidence level of HMI = 73%	

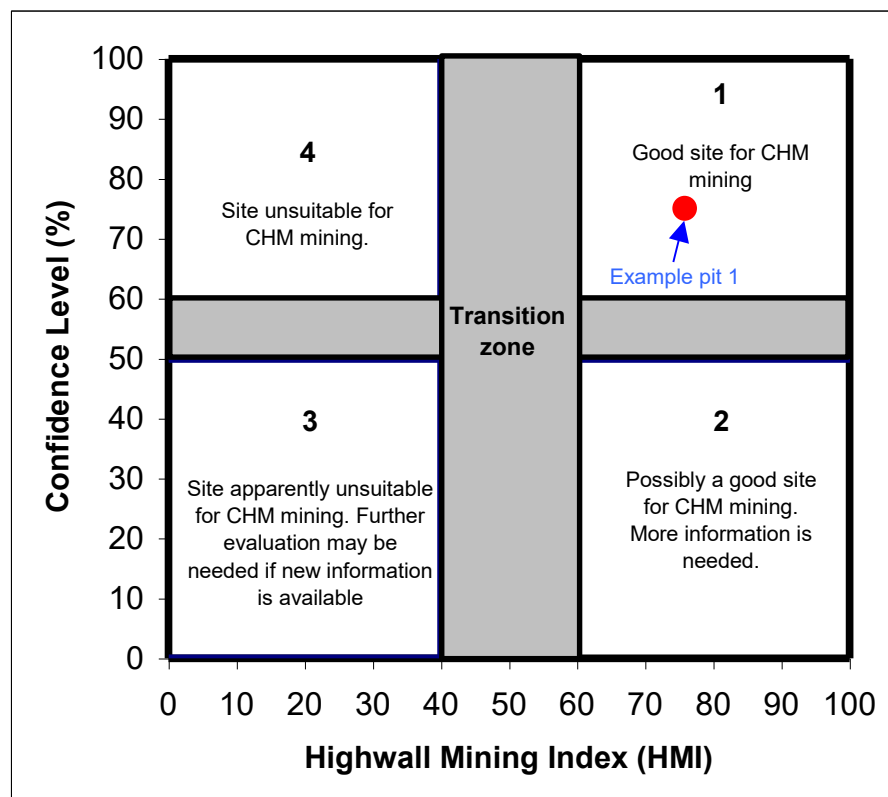


Figure 2.2. Highwall mining index of example site 1.

